MILITARY SPECIFICATION

CLOTH, DENIM, COTTON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for one type of denim cloth.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

V-T-276	-	Thread, Cotton
PPP-P-1134	-	Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

STANDARDS

FEDERAL

FED-STD-4	-	Glossary of Fabric Imperfections
FED-STD-191	-	Textile Test Methods

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes

FSC 8305

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D. C. 20580.)

3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced.

3.2 Material. The cotton shall be carded, drawn and spun into singles yarn.

* 3.3 Color. The color shall be as specified, and shall match the standard sample (see 6.3). The cloth shall have a dyed warp and a natural filling.

* 3.3.1 Labile sulfur. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed cloth shall contain no more labile sulfur, than shown by the standard sample when tested as specified in 4.4. When a standard sample is not available, the dyed cloth shall show no more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.4.

3.3.2 Matching. The color shall match the standard sample under natural (north sky) daylight or artificial daylight at a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin. The warp side shall be used for matching purposes.

3.3.3 Colorfastness. The finished cloth shall show fastness to laundering, perspiration, crocking, and light equal to or better than the standard sample. When no standard sample is established or designated as applicable to colorfastness, the finished cloth shall show "good" fastness to laundering, perspiration, and light after 40 standard fading hours, and shall have a Munsell Value for crocking not lower than 8.5. Fastness shall be determined as specified in 4.4.

3.4 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.4.

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TABLE I. Physical requirements

Weight per square yard (minimum)	Yarns per inch (minimum)		Breaking strength (minimum)		Weave
Ounces	Warp	Filling	Warp	Filling	
			Pounds	Pounds	
5.8	65	42	90	38	1 up, 2 down, right hand twill

3.4.1 Width. The width shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of selvages.

* 3.5 Finish. The warp effect side of the cloth shall be the face.

3.6 pH. The pH value of the water extract of the finished cloth shall be not less than 5.0 nor more than 8.5 when tested as specified in 4.4.

3.7 Nonfibrous material. The cloth shall contain no more than 12.0 percent starch and protein content (including chloroform-soluble and water-soluble material) when tested as specified in 4.4.

3.8 Dimensional stability. Unless otherwise specified (see 6.2), the finished cloth shall not shrink or elongate more than 2.0 percent in either the direction of the warp or of the filling when tested as specified in 4.4. The preshrinking process used shall not be identified by name or trademark either on the cloth, ticket or package.

3.9 Seam efficiency. The finished cloth shall have a seam efficiency of not less than 85 percent when tested as specified in 4.4.

3.10 Length and put-up. Unless otherwise specified (see 6.2), the finished cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put up in flat folds as specified in PPP-P-1134.

3.11 Fiber identification. Each piece shall be labeled or ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.12 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards, when calculated as specified in section 4, shall not exceed the established maximum point value.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.2.1 Component and material inspection. In accordance with 4.1 above, component and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless excluded, amended, modified or qualified in this specification or applicable purchase documents.

* 4.2.2 Examination of the end item. Examination of the end item shall be in accordance with 4.2.2.1 through 4.2.2.4.

* 4.2.2.1 Yard-by-yard examination. Each piece in the sample shall be examined on the warp effect side. When the total yardage in the piece does not exceed 100 yards, the entire yardage in the piece shall be examined. When the total yardage in the piece exceeds 100 yards, only 100 yards shall be examined. All defects as defined in section I of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1 except that only those slubs and knots which exceed the limits shown in figure 1 of FED-STD-4 shall be scored. No folded linear yard from any one piece within the sample shall be penalized more than 4 points. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (pieces) 1/
3200 or less	8
3201 up to and including 10,000	13
10,001 and over	20

1/ No more than one piece shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of pieces in which case all shipping containers shall be present in the sample.

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The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 30 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual pieces exceeds 45 points. If no individual piece exceeds 45 points per 100 square yards, the lot shall be unacceptable with respect to this characteristic. If one piece exceeds 45 points per 100 square yards, a second sample shall be examined only for individual piece quality examination. The lot shall be unacceptable if one or more pieces in the second sample exceeds 45 points per 100 square yards. Point computation for lot quality and individual piece quality shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Contracted width of cloth (inches} \times \text{total yards inspected)}} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension - one point

For defects exceeding 3 inches, but not exceeding 6 inches in any dimension - two points

For defects exceeding 6 inches, but not exceeding 9 inches in any dimension - three points

For defects exceeding 9 inches in any dimension - four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Objectionable odor
Width less than specified
Mottled, streaky or cloudy

4.2.2.2 Examination for length.

4.2.2.2.1 Individual pieces. During the yard-by-yard examination, each piece in the sample shall be examined for length. Any length found to be less than the minimum specified, or more than two yards less than the length marked on the ticket, shall be considered a defect with respect to length. The lot shall be unacceptable if two or more pieces in the sample are defective in respect to length.

4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of pieces in the sample is less than the total of the lengths marked on the tickets.

4.2.2.3 Examination for shade. During the yard-by-yard examination each piece in the sample shall be examined for shade. Any piece in the sample off shade, shaded side to side, side to center, or end to end, shall be cause for rejection of the entire lot represented by the sample.

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4.2.2.4 Examination for identification of preshrinkage process and non-compliance with the Rules and Regulations Under the Textile Fiber Products Identification Act. During the yard-by-yard examination, each piece in the sample shall be examined for these defects. The lot shall be unacceptable if two or more pieces in the sample contain identification of the preshrinkage process by name or trademark on the cloth or ticket, or are not labeled or ticketed in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

* 4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing and marking requirements of section 5 of this specification are complied with.

* 4.4 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table II shall be followed. The physical and chemical values specified in section 3, except when otherwise specified, apply to the results of the determinations made on the sample unit for test purposes as specified in the applicable test method. The sample unit shall be 5 continuous yards full width of the finished cloth. All test reports shall contain the individual values utilized in expressing the final result. The sample size shall be as shown in the list below. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified.

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. Test methods

Characteristic	Requirement paragraph	Test method
Fiber identification:		
Cotton	3.2	1200 1/
Weave	3.4	Visual 2/
Weight	3.4	5041
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Warp	3.4	5100
Filling	3.4	5100

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TABLE II. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method
Presence of Labile sulfur	3.3.1	2020
Colorfastness to:		
Light	3.3.3	5660
Laundering	3.3.3	5610
Perspiration	3.3.3	5680
Crocking	3.3.3	5651
pH	3.6	2811
Nonfibrous material	3.7	2611
Dimensional stability	3.8	5550
Seam efficiency	3.9	5110 3/

1/ Unless otherwise specified a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ One determination shall be made from each sample unit and the result reported as "pass" or "fail".

3/ The needle shall measure 0.040 inch (plus or minus) 0.001 inch across the blade at the eye. The cotton thread shall be type IA1, ticket No. 40, 3 ply in the needle, and ticket No. 60, 3 ply in the loopers, conforming to V-T-276.

* 5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth, put up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

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5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use in the fabrication of uniforms for female personnel of the Department of Defense.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Color required (see 3.3).
(c) Width required (see 3.4.1).
(d) When dimensional stability is not required (see 3.8).
(e) Length required, if other than specified (see 3.10).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids (see 3.1).

* 6.4 Marginal notations. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

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Navy - SA
Air Force - 82

Review activities:

Army - MD
Navy - MC
Air Force - 11

Preparing activity:

Army - GL

Project No. 8305-0102

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-11854F - CLOTH, DENIM, COTTON

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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